

Yuichi HIROI, Ph.D. | 廣井 裕一

Senior Research Scientist, Cluster Metaverse Lab

FORECAST Gotanda WEST 2F, 8-9-5, Nishigotanda, Shinagawa, Tokyo, 141-0031, Japan

 yhiroi.github.io
 y.hiroi@cluster.mu

 scholar.google.com/citations?user=_lCkxzkAAAAJ
 Tokyo, Japan



RESEARCH INTERESTS

Mixed / Augmented Reality (MR/AR), Optical See-Through Head-Mounted Displays (OST-HMDs), Realistic Visual Appearance Reproduction, Vision Augmentation, Metaverse Interaction

WORK EXPERIENCE

Jul. 2023 - Present  Tokyo, Japan	Senior Research Scientist Cluster Metaverse Lab, Cluster Inc.	 Personalized Metaverse Interaction AI Agents in VR/AR
Feb. 2026 - Present  Yokohama, Japan	Visiting Researcher Institute of Science Tokyo, Yuta Itoh Lab.	 AR Displays Vision Augmentation
Jul. 2023 - Mar. 2026  Tokyo, Japan	Visiting Researcher The University of Tokyo, Jun Rekimoto Lab.	 AR Displays Vision Augmentation
Apr. 2022 - Jun. 2023  Tokyo, Japan	JSPS Research Fellow (PD) The University of Tokyo (Advisor : Jun Rekimoto)	 AR Displays Vision Augmentation
Jun. 2021 - Jun. 2023  Kyoto, Japan	Non-full-time Researcher Sony CSL Kyoto (Advisor: Jun Rekimoto)	 AR Displays Neural Rendering
Dec. 2022 - Jan. 2023 Nov. 2019 - Jan. 2020  Otago, New Zealand	Visiting Researcher University of Otago (Advisor : Tobias Langlotz)	 AR Displays Vision Augmentation
Apr. 2020 - Mar. 2022  Yokohama, Japan	JSPS Research Fellow (DC2) Tokyo Institute of Technology (Advisor : Yuta Itoh)	 AR Displays Vision Augmentation
Jan. 2019 - Mar. 2020  Yokohama, Japan	Project Researcher Keio University (Advisor : Maki Sugimoto)	 Collective Visual Sensing
Apr. 2017 - Dec. 2018  Yokosuka, Japan	Full-time Researcher NTT Service Evolution Laboratory	 Object Tracking Conditioned Image Generation
Apr. 2017 - Sep. 2017  Tokyo, Japan	System Engineer (Short-term Contract) NTT DATA Corporation	 Mobile App Development

EDUCATION

Apr. 2019 - Mar. 2022  Tokyo, Japan	Ph. D. of Engineering Tokyo Institute of Technology (Supervisor : Yuta Itoh) Title: "Realistic Appearance Reproduction by Optical See-Through Head-Mounted Display based on Light Measurement and Modulation"
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Apr. 2015 – Mar. 2017
📍 Yokohama, Japan

M.S. of Engineering
Keio University (Supervisor : Maki Sugimoto)

GRANTS

Oct. 2025 – Mar. 2029 (active)	(PI) JST PRESTO “Symbiotic AI Fundamentals” JPMJPR2529 Project: “Infrastructure for Human-AI Collaborative Community Building in the Metaverse”	40.0 M JPY
Apr. 2025 – Mar. 2028 (active)	(PI) JSPS Grant-in-Aid for Scientific Research (B) 25K03173 Project: “Visual Augmentation through Integration of Event-Based Ocular Aberration Measurement and Retinal Projection”	7.7 M JPY
Apr. 2025 – Mar. 2028 (active)	(Co-PI) JSPS Grant-in-Aid for Scientific Research (B) 25K00145 Project: “Tactile Display Using Photo-acoustic Effects Generated by Laser-Induced Cavitation” (PI: Takefumi Hiraki)	3.2 M JPY
Feb. 2024 – Mar. 2027 (active)	(Co-PI) JST ASPIRE for Rising Scientists JPMJAP2327 Project: “Seamless Reality: A Technological Foundation for Merging Cyber-Physical Spaces through Real-World-Oriented Avatars” (PI: Yuta Itoh)	30.6 M JPY
Sep. 2024 – Mar. 2027 (active)	(Co-PI) JSPS KAKENHI for the Promotion of Joint International Research 24KK00187 Project: “Unobtrusive Augmented Reality Visual Guidance with Visual Modality Modulation” (PI: Yuta Itoh)	6.5 M JPY
Apr. 2023 – Mar. 2025	(PI) JSPS KAKENHI for Early-Career Scientists 23K16920 Project: “Spatial Calibration of Head-Mounted Displays Based on Implicit Function Representation of Light Fields Using Deep Learning”	3.5 M JPY
Apr. 2022 – Mar. 2025	(PI) JSPS KAKENHI for JSPS Research Fellow (PD) 22J01340 Project: “Low-latency Vision Augmentation Integrating High-Speed Human and Environmental Measurements with Field-of-View Prediction”	3.4 M JPY
Apr. 2020 – Mar. 2022	(PI) JSPS KAKENHI for JSPS Research Fellow (DC2) 20J14971 Project: “Visual Appearance Reproduction by Optical See-Through Head-Mounted Displays based on Measurement and Modulation of Lights”	1.9 M JPY
Apr. 2019 – Mar. 2020	Tokyo Tech TSUBAME Scholarship for Doctoral Students	0.5 M JPY

AWARDS

International

Aug. 2026	Best Presentation Award, APMAR 2026 Y.-H. Hu, S. Yokoi, Y. Hatada, Y. Hiroi , T. Narumi, T. Hiraki, “The Metaverse as a Scientific Instrument: LUIDA, from Unified Experiment Infrastructure toward Bridging Research and Society”
Oct. 2025	Best Paper Award, IEEE ISEMV 2025 H. Tsutsui, T. Hiraki, Y. Hiroi , S. Hasegawa, “Community Analysis of Social Virtual Reality Based on Large-Scale Log Data”
Aug. 2024	1st Place in SIGGRAPH 2024 Student Research Competition for Undergraduate Work S. Hattori, Y. Hiroi , T. Hiraki, “Measurement of the Imperceptible Threshold for Color Vibration Pairs Selected by using MacAdam Ellipse”
Mar. 2024	Best Paper Honorable Mention, IEEE VR 2024 H. Aoki, T. Tochimoto, Y. Hiroi , Y. Itoh, “Towards Co-operative Beaming Displays: Dual Steering Projectors for Extended Projection Volume and Head Orientation Range”

Mar. 2023	Best Poster Honorable Mention, Augmented Humans 2023 Y. Koike, Y. Hiroi , Y. Itoh, J. Rekimoto. 2023. "Brain-Computer Interface using Directional Auditory Perception"
Mar. 2017	Best Paper 3rd Place, Augmented Human 2017 Y. Hiroi , Y. Itoh, T. Hamasaki, M. Sugimoto, "AdaptiVisor: Assisting Eye Adaptation via Occlusive Optical See-Through Head-Mounted Displays"

Domestic

May. 2025	拡張体験デザイン協会 (DAXX) Good Experience Design Award 2025 R. Kurai, H. Yanagawa, Y. Hiroi , T. Hiraki, "MetaGadget: An Accessible Framework for IoT Integration into Commercial Metaverse Platforms"
Mar. 2025	インタラクティブ発表賞 (PC推薦), インタラクション 2025 沈有方, 土佐凜斗, 畑田裕二, 廣井裕一 , 平木剛史, 苗村健, "HMDユーザの探索行動支援に向けた目立たない両眼相補的色振動による視線誘導", 情報処理学会 インタラクション2025, 2025年
Dec. 2024	特集テーマセッション賞 (ソーシャル・インタラクション), HCGシンポジウム 2024 花鳥諒, 平木剛史, 浦川智弘, 倉井龍太郎, 廣井裕一 , 大山潤爾, "メタバース社会のコミュニケーションにおける感情表現の大規模調査：ソーシャルVRのエモット分析から", 電子情報通信学会 HCGシンポジウム, 2024年
Jun. 2023	2022年 日本の光学研究を代表する成果に選定, 日本光学会 Y. Hiroi , K. Someya, Y. Itoh, "Neural Distortion Fields for Spatial Calibration of Wide Field-of-View Near-Eye Displays", OSA Optics Express, 2022.
Mar. 2019	SIG-MR賞, 日本VR学会複合現実感研究会 浜崎巧, 伊藤勇太, 廣井裕一 , 岩井大輔, 杉本麻樹, "光学透過型ヘッドマウントディスプレイを組み合わせた高ダイナミックレンジなプロジェクションマッピング", 第57回 複合現実感研究会 (SIG-MR), 2019年
Feb. 2014	CG-ARTS賞, 画像情報教育振興協会 廣井裕一 , CG-ARTS主催のエキスパート認定試験で優秀な成績を収める (CGエンジニアエキスパート、画像処理エンジニアエキスパート)

TEACHING EXPERIENCE

Oct. 2026 - Dec. 2026	Lecturer on "Contents Development" (OATW217)
Oct. 2025 - Dec. 2025	University of Tsukuba Delivered a hands-on workshop on video production with Adobe After Effects, complemented by lectures on the underlying principles of digital image processing, color science, and computer graphics for MEng.
Apr. 2020 - Jul. 2020	Teaching Assistant on "Augmented Reality" (CSC.T439)
Apr. 2019 - Jul. 2019	Tokyo Institute of Technology Implemented code, created teaching materials and supported the class for MEng/Ph.D. students to develop applications with full-scratch implementation of AR markers using Processing, OpenGL and OpenCV.
Sep. 2015 - Mar. 2016	Teaching Assistant on "Experiments in Information Engineering II" Keio University Implemented code, created teaching materials and supported the class for Bachelor 3 rd students to develop 3D AR applications with Unity and PlayStation Move Controller.

SKILLS

Technical	Very Experienced: Python, PyTorch, MATLAB, C++, C#, Microsoft Office, Adobe Aftereffects, Adobe Animate, Experienced: Unity, C, Java, JAX, TensorFlow, Processing, Arduino, Optics Design for Head-Mounted Displays and Projectors, WebRTC, Adobe Illustrator, Adobe Photoshop, Adobe InDesign, Blender, Git, Basics: Network Communication (WebRTC), JavaScript, Motion Capture, Raspberry Pi, Robot Operation System (ROS)
Language	Japanese: Native English: Advanced / CEFR C1 (EIKEN English Proficiency Test Grade 1, Mar 2025) Chinese: Pre-Intermediate / CEFR A2 (HSK Level 3, Jul 2024) Korean: Basic / CEFR A1 (TOPIK Level 1, Dec 2025) French, Russian & Arabic : Beginner

PATENTS

US Patent				
Registered	2022	Y. Ishii, K. Hidaka, Y. Tonomura, T. Tokunaga, Y. Hiroi , "Object tracking device, object tracking method, and object tracking program", Nippon Telegraph and Telephone Corp, 2022-07-26.		US11398049
		Y. Hiroi , Y. Ishii, T. Tokunaga, Y. Tonomura, K. Hidaka, "Object tracker, object tracking method, and computer program", Nippon Telegraph and Telephone Corp, 2022-02-22		US11257224
Japanese Patent				
Registered	2026	H. Yanagawa, Y. Hiroi , T. Hiraki, "Information Processing System, Information Processing Methods, and Programs", Cluster Inc., 2026-05-11		No. 7861975
	2021	Y. Hiroi , Y. Ishii, T. Tokunaga, Y. Tonomura, K. Hidaka, "Object tracker, object tracking method, and computer program", Nippon Telegraph and Telephone Corp, 2021-03-10		No. 6850741
	2020	Y. Ishii, K. Hidaka, Y. Tonomura, T. Tokunaga, Y. Hiroi , "Object tracking device, object tracking method, and object tracking program", Nippon Telegraph and Telephone Corp, 2020-11-11.		No. 6793151
Pending	2025	M. Ibara, T. Hiraki, T. Kamegai, S. Yokoi, S. Tokida, Y. Hiroi , "Information Processing Method, Equipment, Program, and System", Cluster Inc., 2025-08-28		Appl. No. 2025-141831
	2024	Y. Hiroi , T. Hiraki, "Information Processing Equipment, Information Processing Methods, and Programs", Cluster Inc., 2024-03-12		Appl. No. 2024-037967

SERVICE

Editorial Service				
Journal Editor	2025 - Present	Journal Editor Committee, Virtual Reality Society of Japan (VRSJ)		
Guest Editor	2026	Special Issue on "AI and VR", Journal of VRSJ		
Conference / Workshop Organization	2027	General Chair , APMAR 2027		
	2026	General Chair , APMAR 2026		
		Seminar Organizer - NII Shonan Meeting No. 242 "Seamless Reality"		
		Workshop Organizer - "4 th & 5 th Workshop on Seamless Reality : AR Technologies for Seamless Perception and Cognition between Cyber and Physical Spaces (WSR)", IEEE VR 2026 / ISMAR 2026		
	2025	Program Chair , APMAR 2025		
		Workshop Organizer - "2 nd & 3 rd Workshop on Seamless Reality : AR Technologies for Seamless Perception and Cognition between Cyber and Physical Spaces (WSR)", IEEE VR 2025 / ISMAR 2025		
	2024	Workshop Organizer - "1 st Workshop on Seamless Reality : AR Technologies for Seamless Perception and Cognition between Cyber and Physical Spaces (WSR)", IEEE VR 2024		

Program Committees (PC)	2027	IEEE VR
	2026	IEEE VR, IEEE ISMAR, ACM VRST, ACM ICMI, ICAT-EGVE, EuroXR
	2025	IEEE VR, IEEE ISMAR, ICAT-EGVE
	2024	IEEE ISMAR, ICAT-EGVE, APMAR
Journal Reviewer		
International	2026	IEEE TVCG x3, OPTICA Optics Express x2, Optics Letters x1
	2025	IEEE TVCG x1, OPTICA Optics Express x4, Optics Letters x1, Applied Optics x1
	2024	IEEE TVCG x2, Elsevier Displays x1
	2023	OPTICA Applied Optics x1
	2021	OPTICA Optics Express x1
Domestic	2023	Journal of the Virtual Reality Society of Japan (VRSJ) x1
Conference Reviewer	2026	IEEE VR x8 (PC), IEEE ISMAR x7 (PC), ACM VRST x6 (PC), ACM ICMI x4 (PC), EuroXR x4 (PC), ICAT-EGVE x2 (PC), SIGGRAPH x1, SIGGRAPH Asia E-tech x2
	2025	IEEE VR x9 (PC), IEEE ISMAR x9 (PC), ICAT-EGVE x2 (PC)
	2024	IEEE ISMAR x5 (PC), ICAT-EGVE x3 (PC), ACM VRST x3, IEEE VR x2, ACM CHI x1, ACM SUI x1, APMAR x1 (PC), SIGGRAPH Asia E-Tech x1
	2023	IEEE ISMAR x3, IEEE VR x1, ACM SUI x1
Academic Society Memberships	Committee - Domestic: VRSJ Special Interests Group of Mixed Reality (SIG-MR, 2023 - Present)	
	Member - International: IEEE, ACM Domestic: VRSJ	
Others	2026	Session Chair, IEEE AIxVR
	2025	Session Chair, IEEE VR, IMID
	2024	Session Chair, IEEE VR
	2022	Session Chair, ACM VRST

PUBLICATIONS AND PRESENTATION LIST

International Journals & Magazines (Peer-Reviewed)	(in review) Yuichi Hiroi , Yuki Imai, Hikari Yanagawa, Takefumi Hiraki, "Large-scale, Longitudinal Field Study of AI-agent-User Interactions in Commercial Metaverse", <i>Frontiers in Virtual Reality</i> .		
	(in review) Shingo Hattori, Yuta Itoh, Takefumi Hiraki, Yuichi Hiroi , "A Cone-opponent Amplitude Budget for Imperceptible Color Vibration on Head-mounted Displays", <i>Optics Express</i> .		
	(in review) Chun Wei Ooi, Yuichi Hiroi , Takefumi Hiraki, Yuta Itoh, "Toward Real-time Photochromic Occlusion: Characterizing a Fast-response Photochromic Compound for Near-eye Displays", <i>Optics Express</i> .		
	2026	(to appear) Hikari Yanagawa, Yuichi Hiroi, Takefumi Hiraki, "DELUGE: Decomposed Entropy-coded Live Unstructured Geometry Exchange for Real-time Particle Streaming", <i>IEEE Transaction on Visualization and Computer Graphics (TVCG)</i>, 2026. (Also presented at IEEE ISMAR 2026) (to appear) Yong-Hao Hu, Sotaro Yokoi, Yuji Hatada, Yuichi Hiroi, Takuji Narumi, Takefumi Hiraki, "LUIDA: Large-scale Unified Infrastructure for Digital Assessments based on Commercial Metaverse Platform", <i>Frontiers in Virtual Reality</i>, 2026. - Michikuni Eguchi, Tomohiro Hayase, Yuichi Hiroi, Takefumi Hiraki, "Cluster Haptic Texture Database: Haptic Texture Database with Variety in Velocity and Direction of Sliding Contacts", <i>Nature Scientific Data</i>, vol. 13, 756, 2026. - Jonas Weigand, Christian Eichhorn, Yuichi Hiroi, Yuta Itoh, "BeamStellar: Beaming Display with Low-Latency, 6-DoF Glasses Tracking via Spatio-Temporal LED Encoding", <i>IEEE Transactions on Visualization and Computer Graphics</i>, vol. 32, no. 5, pp. 3222-3228, 2026. (Also presented at IEEE VR 2026)	

- 2026 5. Ryutaro Kurai, Takefumi Hiraki, **Yuichi Hiroi**, Yutaro Hirao, Monica Perusquia-Hernandez, Hideaki Uchiyama, Kiyoshi Kiyokawa, "MetaProxy: Bridging Agents and Environments in Commercial Metaverse Platforms through Remote Intermediary Monitoring", *IEICE Transactions on Information and Systems*, vol. E109.D, no. 6, pp. 870-878, 2026.
- 2025 6. **Yuichi Hiroi**, Yuji Hatada, Takefumi Hiraki, "Cross-Reality Lifestyle: Integrating Physical and Virtual Lives through Multi-Platform Metaverse", *IEEE Pervasive Computing*, vol. 24, no. 4, pp. 19-28, 2025.
7. Ryutaro Kurai, Hikari Yanagawa, **Yuichi Hiroi**, Takefumi Hiraki, "MetaGadget: An Accessible Framework for IoT Integration into Commercial Metaverse Platforms", *IEEE Pervasive Computing*, vol. 24, no. 4, pp. 63-73, 2025.
8. Ryutaro Kurai, Takefumi Hiraki, **Yuichi Hiroi**, Yutaro Hirao, Monica Perusquia-Hernandez, Hideaki Uchiyama, Kiyoshi Kiyokawa, "MagicCraft: Natural Language-Driven Generation of Dynamic and Interactive 3D Objects for Commercial Metaverse Platforms", *IEEE Access*, vol. 13, pp. 132459-132474, 2025.
9. Rinto Tosa, Shingo Hattori, **Yuichi Hiroi**, Yuta Itoh, Takefumi Hiraki, "ChromaGazer: Unobtrusive Visual Modulation using Imperceptible Color Vibration for Visual Guidance", *IEEE Transaction on Visualization and Computer Graphics (TVCG)*, vol. 31, no. 5, pp. 3450-3458, 2025 (Also presented at IEEE VR 2025).
10. Ryutaro Kurai, Takefumi Hiraki, **Yuichi Hiroi**, Yutaro Hirao, Monica Perusquia-Hernandez, Hideaki Uchiyama and Kiyoshi Kiyokawa, "MagicItem: Dynamic Behavior Design of Virtual Objects with Large Language Models in a Commercial Metaverse Platform", *IEEE Access*, vol. 13, pp. 19132-19143, 2025.
- 2024 11. Rina Nagano, Takehiro Kinoshita, Shingo Hattori, **Yuichi Hiroi**, Yuta Itoh, Takefumi Hiraki, "HaptoFloater: A Low-latency Visuo-Haptic Mid-Air Display by Embedding Imperceptible Color-Vibration Signals", *IEEE Transaction on Visualization and Computer Graphics (TVCG)*, vol. 30, no. 11, pp. 7463-7472, 2024 (Also presented at IEEE ISMAR 2024)
12. Hiroto Aoki, Takumi Tochimoto, **Yuichi Hiroi**, Yuta Itoh, "Towards Co-operative Beaming Displays: Dual Steering Projectors for Extended Projection Volume and Head Orientation Range," *IEEE Transaction on Visualization and Computer Graphics (TVCG)*, vol. 30, no. 5, pp. 2309-2318, 2024 (Also presented at IEEE VR 2024)
13. **Yuichi Hiroi**, Takefumi Hiraki, Yuta Itoh, "StainedSweeper: Compact, Variable-Intensity Light-Attenuation Display with Sweeping Tunable Retarders", *IEEE Transaction on Visualization and Computer Graphics (TVCG)*, vol. 30, no. 5, pp. 2682-2692, 2024 (Also presented at IEEE VR 2024)
- 2023 14. **Yuichi Hiroi**, Akira Watanabe, Yuri Mikawa, Yuta Itoh, "Low-Latency Beaming Display: Implementation of Wearable, 133 μ s Motion-to-Photon Latency Near-eye Display", *IEEE Transaction on Visualization and Computer Graphics (TVCG)*, vol. 29, no. 11, pp. 4761-4771, 2023 (Also presented at IEEE ISMAR 2023)
- 2022 15. **Yuichi Hiroi**, Kiyosato Someya, Yuta Itoh, "Neural Distortion Fields for Spatial Calibration of Wide Field-of-View Near-Eye Displays", *Optics Express*, Vol. 30, Issue 22, pp. 40628-40644, 2022.
- 2021 16. **Yuichi Hiroi**, Takumi Kaminokado, Shunsuke Ono, Yuta Itoh, "Focal Surface Occlusion", *Optics Express*, Vol. 29, Issue 22, pp. 36581-36597, 2021.
- 2020 17. Takumi Kaminokado, **Yuichi Hiroi**, Yuta Itoh, "StainedView: Variable-Intensity Light Attenuation Display with Cascaded Spatial Color Filtering for Improved Color Fidelity," *IEEE Transactions on Visualization and Computer Graphics (TVCG)* 26(6): pp. 3576-3856, 2020 (Also presented at IEEE ISMAR 2020)
- 2018 18. Takumi Hamasaki, Yuta Itoh, **Yuichi Hiroi**, Daisuke Iwai, Maki Sugimoto, "HySAR: Hybrid Material Rendering by an Optical See-Through Head-Mounted Display with Spatial Augmented Reality Projection". *IEEE Transaction on Visualization and Computer Graphics (TVCG)* 24(4): pp. 1457-1466, 2018 (Also presented at IEEE VR 2018).

**International
Conference
Proceedings
(Peer-Reviewed)**

- 2026 1. (to appear) Michikuni Eguchi, **Yuichi Hiroi**, Takefumi Hiraki, "HaptoFlow: High-Fidelity Real-Time Vibrotactile Generation via Flow Matching for Virtual Reality", In *Proceedings of International Symposium on Mixed and Augmented Reality (ISMAR 2026)*, Bari, Italy, Oct. 5-9, 2026.
2. Satomi Tokida, Yong-Hao Hu, Yuta Itoh, Jean-Marie Normand, Rebecca Fribourg, Jean-Philippe Riviere, **Yuichi Hiroi**, Takefumi Hiraki, Yoshio Ishiguro, "ImmersiveDMT: A VR System for Expressive Body Movement and Stress Reduction through Dance Movement Therapy", In *Proceedings of IEEE International Conference of Virtual Reality 2026 (IEEE VR 2026)*, 2026.
3. **Yuichi Hiroi**, Yuki Imai, Hikari Yanagawa, Takefumi Hiraki, "Large-scale, Longitudinal Field Study of AI-agent-User Interactions in Commercial Metaverse", In *Proceeding of IEEE International Conference on Artificial Intelligence and eXtended and Virtual Reality (AIxVR 2026)*, Osaka, Japan, 11 pages, Jan 26-28, 2026.
- 2025 4. Masahiro Nara, Ryusuke Miyazaki, **Yuichi Hiroi**, Takefumi Hiraki, Yuta Itoh, Shio Miyafuji, "Saccadance: Saccade-Aware Pattern Embedding for Gaze Guidance on High-Speed Displays", In *Proceedings of International Conference of Virtual Reality Software Technology (VRST 2025)*, Montreal, Canada, 11 pages, Nov. 11-14, 2025.

- 2025 5. Hiroto Tsutsui, Takefumi Hiraki, **Yuichi Hiroi** and Shoichi Hasegawa, "Community Analysis of Social Virtual Reality Based on Large-Scale Log Data", 2nd IEEE International Symposium on Emerging Metaverse (ISEMV 2025), Honolulu, USA, 9 pages, Oct. 19-20, 2025.
6. Masanori Ibara, **Yuichi Hiroi**, Takushi Kamegai and Takefumi Hiraki. "Implementation and Application of Multi-Format 3D Data Integration in a Cross-Device Commercial Metaverse Platform", Proceedings of the 2nd IEEE International Symposium on Emerging Metaverse (ISEMV 2025), Honolulu, USA, Oct. 19-20, 10 pages, 2025.
7. Masanori Ibara, **Yuichi Hiroi**, Takushi Kamegai, Yusuke Masubuchi, Kazuki Matsutani, Megumi Zaizen, Junya Morita, and Takefumi Hiraki, "Weather Synchronization in Digital Twin Environments for Shared VR Experience Using Commercial Metaverse Platforms," Proceedings of the 2nd IEEE International Symposium on Emerging Metaverse (ISEMV 2025), Honolulu, USA, Oct. 19-20, 4 pages, 2025.
8. Takekazu Kitagishi, Chun Wei Ooi, **Yuichi Hiroi**, Jun Rekimoto. "Pinching Visuo-haptic Display: Investigating Cross-Modal Effects of Visual Textures on Electrostatic Cloth Tactile Sensations", 27th ACM International Conference on Multimodal Interaction (ICMI 2025), Canberra, Australia, Oct. 13-17, 2025.
9. Hikari Yanagawa, **Yuichi Hiroi**, Satomi Tokida, Yuji Hatada, Takefumi Hiraki, "Navigation Pixie: Implementation and Empirical Study Toward On-demand Navigation Agents in Commercial Metaverse", In Proceedings of International Symposium on Mixed and Augmented Reality (ISMAR 2025), Daejeon, Korea, Oct. 8-12, 2025.
10. Youfang Shen, Rinto Tosa, Yuji Hatada, **Yuichi Hiroi**, Takefumi Hiraki, Takeshi Naemura, "ChromaGazer-HMD: Visual Modulation using Unobtrusive Color Vibration for Gaze Guidance with Head-Mounted Displays", Augmented Humans 25, Abu Dhabi, UAE, 2025.
11. Yuta Itoh, Tomoya Nakamura, **Yuichi Hiroi**, Kaan Aksit, "Slim Light-Receiving Glasses for Wide Head Orientation-Capable Beaming Displays using Diffractive Waveguide", In Proceedings of IEEE International Conference of Virtual Reality 2025 (IEEE VR 2025), pp. 351-358, Saint-Malo, France, Mar. 8-12, 2025.
- 2024 12. **Yuichi Hiroi**, Takefumi Hiraki, Yuta Itoh, "FactoredSweeper: Optical See-Through Display Integrating Light Attenuation and Addition with Single Spatial Light Modulator", In Proceedings of International Symposium on Mixed and Augmented Reality (ISMAR 2024), pp. 61-70, Seattle, USA, Oct. 21-25, 2024.
- 2023 13. Hiroto Aoki, **Yuichi Hiroi**, Yuta Itoh, Jun Rekimoto, "Retinal Homing Display: Head-Tracking Auto-stereoscopic Retinal Projection Display", In Proceedings of International Conference of Virtual Reality Software Technology (VRST 2023), 13:1-13:10, Christchurch, New Zealand, 2023
14. Takekazu Kitagishi, **Yuichi Hiroi**, Yuna Watanabe, Yuta Itoh, Jun Rekimoto, "Telexile: End-to-end Remote Transmission of Fabric Tactile Sensation", In Proceedings of the ACM Symposium on User Interface Software and Technology 2023 (UIST 2023), 67:1-67:10, San Francisco, USA, 2023.
15. Chun Wei Ooi, **Yuichi Hiroi**, Yuta Itoh, "A Compact Photochromic Occlusion Capable See-through Display with Holographic Lenses", In Proceedings of IEEE International Conference of Virtual Reality 2023 (IEEE VR 2023), pp. 237-242, Shanghai, China, 2023.
- 2022 16. **Yuichi Hiroi**, Yuta Itoh, Jun Rekimoto, "NeARportation: A Remote Real-time Neural Rendering Framework", In Proceedings of International Conference of Virtual Reality Software Technology (VRST 2022), 23:1-23:5, Tsukuba, Japan, 2022.
17. Zhang Zhibin, **Yuichi Hiroi**, Yuta Itoh, "Towards Spatial Airflow Interaction: Schlieren Imaging for Augmented Reality", In Proceedings of International Symposium on Mixed and Augmented Reality (ISMAR 2022), pp. 215-223, Singapore, Singapore, 2022.
- 2021 18. Takumi Tochimoto, **Yuichi Hiroi**, Yuta Itoh, "CircadianVisor: Image Presentation With an Optical SeeThrough Display in Consideration of Circadian Illuminance", In Proceedings of the Augmented Humans International Conference (AHs 2021), pp. 66-76, 2021.
- 2020 19. **Yuichi Hiroi**, Takumi Kaminokado, Atsushi Mori, Yuta Itoh, "DehazeGlasses: Optical Dehazing with an Occlusion Capable See-Through Display," In Proceedings of the Augmented Humans International Conference (AHs 2020), 3:1-3:11, Virtual, 2020.
- 2017 20. **Yuichi Hiroi**, Yuta Itoh, Takumi Hamasaki, Maki Sugimoto, "AdaptiVisor: Assisting Eye Adaptation via Occlusive Optical See-Through Head-Mounted Displays," The 8th Augmented Human International Conference (AH 2017), 9:1-9:9, Silicon Valley, USA, Mar. 16-18, 2017.

**International
Conference
Workshop
Proceedings
(Peer-Reviewed)**

- 2026 1. (to appear) Norichika Matsuno, Yutaro Hirao, **Yuichi Hiroi**, Takefumi Hiraki, "SnapGrid: A Predictable Snap Rotation Technique with a Visible Phase Cue for Cybersickness Mitigation in Virtual Reality Experiences", 5th Workshop on Seamless Reality: AR Technologies for Seamless Perception and Cognition between Cyber and Physical Spaces (5th WSR workshop, IEEE ISMAR-Adjunct), 2026.
2. (to appear) Masanori Ibara, Yutaro Hirao, **Yuichi Hiroi**, Takefumi Hiraki, "AdaptLBE: Toward Co-optimizing Real-Object Placement and Redirected Walking to Host an Existing VR World in an Ordinary Room", 5th Workshop on Seamless Reality: AR Technologies for Seamless Perception and Cognition between Cyber and Physical Spaces (5th WSR workshop, IEEE ISMAR-Adjunct), 2026.

- 2026 3. Satomi Tokida, **Yuichi Hiroi**, Miyabi Uesugi, Atsushi Shimono, Haruyuki Suzuki, Shouma Abe, Takefumi Hiraki, "Analyzing User Behaviors in Metaverse Brand Events that Drive Real-World Store Visit Intentions", 4th Workshop on Seamless Reality: AR Technologies for Seamless Perception and Cognition between Cyber and Physical Spaces (4th WSR workshop, IEEEVR-Adjunct), 2026.
 4. Sotaro Yokoi*, Satomi Tokida*, **Yuichi Hiroi**, Takefumi Hiraki, "Verselyzer: A Statistical Analysis Interface for Metaverse Branding Effectiveness Assessment by Non-Experts", 4th Workshop on Seamless Reality: AR Technologies for Seamless Perception and Cognition between Cyber and Physical Spaces (4th WSR workshop, IEEEVR-Adjunct), 2026 (*equally contributed).
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Yuta Itoh

Associate Professor, Institute of Science Tokyo
4259 Nagatsuta-cho, Midori-ku, Yokohama, Kanagawa,
226-8501, Japan
ito@comp.isct.ac.jp

Maki Sugimoto

Professor, Keio University
3-14-1, Hiyoshi, Yokohama, Kanagawa, Japan
223-8522
sugimoto@ics.keio.ac.jp

Daisuke Iwai

Professor, Osaka University
1-3, Machikaneyama, Toyonaka, Osaka, Japan
560-8531
daisuke.iwai@sys.es.osaka-u.ac.jp